

General Electrical Engineering Data Manual

Page 1 of 1 | PowerCalc Engineering Manual Suite

1. Fundamental Formulas & Ohm's Law

Direct Current: $V = I * R$, $P = V * I = I^2 * R = V^2 / R$

Single-Phase AC: $P = V * I * \cos(\phi)$, $S = V * I$, $Q = V * I * \sin(\phi)$

Three-Phase AC: $P = \sqrt{3} * V_L * I_L * \cos(\phi)$, $S = \sqrt{3} * V_L * I_L$

Power factor: $\cos(\phi) = \text{Active Power (kW)} / \text{Apparent Power (kVA)}$

2. Conductor Voltage Drop Calculation (IEC & NEC)

Single-Phase: $\Delta V = [2 * L * I * (R * \cos(\phi) + X * \sin(\phi))] / 1000$

Three-Phase: $\Delta V = [\sqrt{3} * L * I * (R * \cos(\phi) + X * \sin(\phi))] / 1000$

Standard permissible drop: Lighting $\leq 3\%$, Power/Feeders $\leq 5\%$ total.

3. International Wiring Color Identification

IEC Standard: Phase 1=Brown, Phase 2=Black, Phase 3=Grey, Neutral=Blue, Earth=Green/Yellow

NEC (120/208V): Phase A=Black, Phase B=Red, Phase C=Blue, Neutral=White, Ground=Green/Bare

NEC (277/480V): Phase A=Brown, Phase B=Orange, Phase C=Yellow, Neutral=Grey, Ground=Green